A Rigorous Lower Bound and Finiteness Proof for a Two-Colour Finite-Sums Ramsey Number

Abstract

Let $F(k)$ be the least integer N such that every two-colouring of $\{1, \dots, N\}$ contains a k-element set A for which all non-empty subset sums of A are defined inside $\{1, \dots, N\}$ and have the same colour. We prove, in ZFC, that

$$4^{k-1} \le F(k) < \infty.$$

The lower bound is obtained from the colouring $n \mapsto \nu_2(n) \pmod 2$. The upper bound is qualitative: it follows from Hindman's finite-sums theorem, proved here by the Galvin–Glazer ultrafilter argument, followed by a compactness extraction.

1 Definition and statement of the result

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. If A is a finite set of positive integers, define

$$\mathrm{FS}(A) := \left\{ \sum_{a \in S} a : \varnothing \ne S \subseteq A \right\}.$$

Definition 1.1. *For $k \ge 1$, let $F(k)$ be the least integer N, if it exists, such that every colouring*

$$\chi : \{1, \dots, N\} \to \{0, 1\}$$

contains a k-element set $A \subseteq \mathbb{N}$ satisfying

$$\mathrm{FS}(A) \subseteq \{1, \dots, N\}$$

and such that χ is constant on $\mathrm{FS}(A)$*.*

The condition $\mathrm{FS}(A) \subseteq \{1, \dots, N\}$ is essential: the colouring is only defined up to N. Equivalently, since all elements of A are positive,

$$\mathrm{FS}(A) \subseteq \{1, \dots, N\} \quad \Longleftrightarrow \quad \sum_{a \in A} a \le N.$$

The main result proved in this note is the following.

Theorem 1.2. *For every integer $k \ge 1$, the number $F(k)$ exists and satisfies*

$$4^{k-1} \le F(k) < \infty.$$

The result is an estimate rather than an exact evaluation. The lower bound is explicit and elementary. The finiteness proof is non-quantitative and uses Hindman's theorem; it gives no practical numerical upper bound.

2 The explicit lower bound

Let $\nu_2(n)$ denote the exponent of 2 in the prime factorisation of n. Thus $\nu_2(n) = t$ means that $2^t \mid n$ but $2^{t+1} \nmid n$.

We shall use the two-colouring

$$\chi(n) := \nu_2(n) \pmod 2.$$

The lower bound follows from a structural lemma about finite-sums sets under this colouring.

Lemma 2.1. *Let A be a k-element set of positive integers. Suppose that every non-empty subset sum of A has even 2-adic valuation; that is,*

$$\nu_2\left(\sum_{a\in S} a\right) \equiv 0 \pmod 2$$

for every non-empty $S \subseteq A$. Then

$$\sum_{a\in A} a \geq 4^{k-1}.$$

Proof. We argue by induction on k.

For $k = 1$, the claim is immediate, since a positive integer is at least $1 = 4^0$.

Assume $k \geq 2$, and assume the lemma has been proved for all sets of size $k-1$. Let A be a k-element set satisfying the hypothesis.

First remove a harmless common factor. Since every singleton sum belongs to $\mathrm{FS}(A)$, each $a \in A$ has even 2-adic valuation. Therefore every $a \in A$ is of the form $4^{e_a}u_a$ with u_a odd. Let

$$e := \min_{a\in A} e_a.$$

Set

$$A_0 := \{a/4^e : a \in A\}.$$

Then A_0 still has k elements. Moreover, for every non-empty $S \subseteq A$,

$$\nu_2\left(\sum_{a\in S} \frac{a}{4^e}\right) = \nu_2\left(\sum_{a\in S} a\right) - 2e,$$

so every non-empty subset sum of A_0 also has even 2-adic valuation. Finally, by the choice of e, at least one element of A_0 is odd, and

$$\sum_{a\in A} a = 4^e \sum_{b\in A_0} b \geq \sum_{b\in A_0} b.$$

Thus it is enough to prove the desired bound in the case where A contains at least one odd element.

Assume from now on that A contains at least one odd element. Since every singleton has even 2-adic valuation, every non-odd element of A is divisible by 4.

There cannot be three odd elements in A. Indeed, among any three odd integers, two are congruent modulo 4. The sum of those two integers is then congruent to 2 modulo 4, and therefore has 2-adic valuation exactly 1, contradicting the hypothesis.

So A contains either one or two odd elements.

Case 1: exactly one odd element. Write

$$A = \{u\} \cup 4B,$$

where u is odd and B is a $(k-1)$-element set of positive integers. For every non-empty $T \subseteq B$,

$$4\sum_{b\in T} b$$

is a non-empty subset sum of A, hence has even 2-adic valuation. Since it is divisible by 4, dividing by 4 subtracts 2 from its valuation. Therefore

$$\nu_2\left(\sum_{b\in T} b\right)$$

is even. By the induction hypothesis,

$$\sum_{b\in B} b \geq 4^{k-2}.$$

Consequently,

$$\sum_{a\in A} a = u + 4\sum_{b\in B} b \geq 1 + 4\cdot 4^{k-2} > 4^{k-1}.$$

Case 2: exactly two odd elements. Let these two odd elements be u and v. Since $u+v$ is a non-empty subset sum of A, its 2-adic valuation is even. Also $u+v$ is even. Hence $\nu_2(u+v) \geq 2$, so

$$4 \mid u+v.$$

Define

$$B := \left\{\frac{u+v}{4}\right\} \cup \left\{\frac{x}{4} : x \in A \setminus \{u,v\}\right\}.$$

We first check that B really has $k-1$ distinct elements. The elements $x/4$ with $x \in A \setminus \{u,v\}$ are distinct because the corresponding x are distinct. It remains only to check that $(u+v)/4$ is not equal to one of them. If $(u+v)/4 = x/4$ for some $x \in A \setminus \{u,v\}$, then $x = u+v$, and the subset sum

$$u+v+x = 2(u+v)$$

would have 2-adic valuation $\nu_2(u+v)+1$, which is odd because $\nu_2(u+v)$ is even. This contradicts the hypothesis. Thus B has exactly $k-1$ elements.

Now we verify that every non-empty subset sum of B has even 2-adic valuation. Such a subset sum has one of two forms. Either it is

$$\sum_{x\in T} \frac{x}{4}$$

for some non-empty $T \subseteq A \setminus \{u,v\}$, or it is

$$\frac{u+v}{4} + \sum_{x\in T} \frac{x}{4}$$

for some possibly empty $T \subseteq A \setminus \{u,v\}$.

In the first case, multiplying by 4 gives the subset sum $\sum_{x\in T} x$ of A. This has even 2-adic valuation and is divisible by 4, so after division by 4 the valuation remains even.

In the second case, multiplying by 4 gives the subset sum

$$u+v+\sum_{x\in T} x$$

of A. This again has even 2-adic valuation and is divisible by 4, since $u+v$ and every $x \in A\setminus\{u,v\}$ are divisible by 4. Thus division by 4 again preserves the parity of the valuation.

Therefore B satisfies the induction hypothesis. Hence

$$\sum_{b \in B} b \geq 4^{k-2}.$$

But by construction,

$$\sum_{a \in A} a = 4 \sum_{b \in B} b.$$

Therefore

$$\sum_{a \in A} a \geq 4 \cdot 4^{k-2} = 4^{k-1}.$$

The induction is complete. $\square$

Corollary 2.2. *For every $k \geq 1$,*

$$F(k) \geq 4^{k-1}.$$

Proof. Colour $\mathbb{N}$ by

$$\chi(n) = \nu_2(n) \pmod 2.$$

Suppose A is a k-element set whose non-empty subset sums are monochromatic under this colouring. Let

$$t := \min_{a \in A} \nu_2(a).$$

All singleton sums have the same colour, so each $\nu_2(a)$ has the same parity. Since t is one of these valuations, every $\nu_2(a) - t$ is even. Thus

$$A' := \{a/2^t : a \in A\}$$

is a k-element set of positive integers.

For every non-empty $S \subseteq A$,

$$\nu_2\left(\sum_{a \in S} \frac{a}{2^t}\right) = \nu_2\left(\sum_{a \in S} a\right) - t.$$

The first term on the right has the common colour parity of all subset sums, and t has that same parity because singleton sums are among the subset sums. Hence the difference is even. By Lemma 2.1,

$$\sum_{a \in A'} a \geq 4^{k-1}.$$

Therefore

$$\sum_{a \in A} a = 2^t \sum_{a \in A'} a \geq 4^{k-1}.$$

Now restrict this colouring to $\{1, \ldots, 4^{k-1} - 1\}$. If a valid k-element set A existed there, then $\mathrm{FS}(A)$ would have to be contained in that interval, so in particular

$$\sum_{a \in A} a \leq 4^{k-1} - 1,$$

contradicting the inequality just proved. Hence no such set is forced below 4^{k-1}, and so $F(k) \geq 4^{k-1}$. $\square$

3 Ultrafilters and the Galvin–Glazer proof of Hindman's theorem

The lower bound above is elementary. To prove finiteness of $F(k)$, we use Hindman's finite-sums theorem. For completeness, we give a self-contained proof of the needed infinite theorem using the standard ultrafilter method.

3.1 The semigroup $\beta\mathbb{N}$

An *ultrafilter* on $\mathbb{N}$ is a collection p of subsets of $\mathbb{N}$ satisfying:

(i) $\varnothing \notin p$;

(ii) if $A, B \in p$, then $A \cap B \in p$;

(iii) if $A \in p$ and $A \subseteq B \subseteq \mathbb{N}$, then $B \in p$;

(iv) for every $A \subseteq \mathbb{N}$, exactly one of A and $\mathbb{N} \setminus A$ lies in p.

For $n \in \mathbb{N}$, the *principal ultrafilter at n* is

$$p_n := \{A \subseteq \mathbb{N} : n \in A\}.$$

An ultrafilter is *non-principal* if it is not principal at any $n \in \mathbb{N}$.

Let $\beta\mathbb{N}$ denote the set of all ultrafilters on $\mathbb{N}$. For $A \subseteq \mathbb{N}$, write

$$\overline{A} := \{p \in \beta\mathbb{N} : A \in p\}.$$

The sets $\overline{A}$ form the usual clopen basis for the topology on $\beta\mathbb{N}$: indeed,

$$\beta\mathbb{N} \setminus \overline{A} = \overline{\mathbb{N} \setminus A}, \qquad \overline{A} \cap \overline{B} = \overline{A \cap B}.$$

Lemma 3.1 (Ultrafilter extension)**.** *Let $\mathcal{F}$ be a family of subsets of $\mathbb{N}$ with the finite-intersection property. Then there exists an ultrafilter p on $\mathbb{N}$ such that $\mathcal{F} \subseteq p$.*

Proof. Let $\mathcal{G}$ be the filter generated by $\mathcal{F}$; explicitly, $B \in \mathcal{G}$ if and only if B contains an intersection of finitely many members of $\mathcal{F}$. The finite-intersection property ensures that $\varnothing \notin \mathcal{G}$.

Consider the set of all proper filters on $\mathbb{N}$ containing $\mathcal{G}$, ordered by inclusion. The union of any chain of such filters is again a proper filter containing $\mathcal{G}$. Hence, by Zorn's lemma, there is a maximal proper filter p containing $\mathcal{G}$.

We prove that p is an ultrafilter. Let $A \subseteq \mathbb{N}$. If $A \in p$, there is nothing to prove. Suppose $A \notin p$. If $\mathbb{N} \setminus A \notin p$, then every $B \in p$ must meet A; otherwise $B \subseteq \mathbb{N} \setminus A$ would force $\mathbb{N} \setminus A \in p$ by upward closure. Hence the family $p \cup \{A\}$ has the finite-intersection property and generates a proper filter strictly larger than p, contradicting maximality. Therefore $\mathbb{N} \setminus A \in p$. Thus exactly one of A and $\mathbb{N} \setminus A$ lies in p, so p is an ultrafilter. $\square$

Proposition 3.2. *With this topology, $\beta\mathbb{N}$ is compact and Hausdorff.*

Proof. First, $\beta\mathbb{N}$ is Hausdorff. If $p \neq q$, then some $A \subseteq \mathbb{N}$ belongs to one of p, q but not the other. Say $A \in p$ and $A \notin q$. Then $\mathbb{N} \setminus A \in q$, so $p \in \overline{A}$, $q \in \overline{\mathbb{N} \setminus A}$, and the two basic open sets $\overline{A}$ and $\overline{\mathbb{N} \setminus A}$ are disjoint.

It remains to prove compactness. It is enough to show that every cover of $\beta\mathbb{N}$ by basic clopen sets has a finite subcover, because every open cover has a refinement by basic clopen sets.

Suppose $\{\overline{A_i} : i \in I\}$ is a basic clopen cover with no finite subcover. Then for every finite $J \subseteq I$, the set

$$\mathbb{N} \setminus \bigcup_{j \in J} A_j = \bigcap_{j \in J} (\mathbb{N} \setminus A_j)$$

is non-empty. Indeed, if it were empty, then $\bigcup_{j\in J} A_j = \mathbb{N}$; since an ultrafilter containing a finite union contains at least one member of that finite union, the sets $\overline{A_j}$ with $j \in J$ would already cover $\beta\mathbb{N}$, contrary to the assumption. Therefore the family $\{\mathbb{N} \setminus A_i : i \in I\}$ has the finite-intersection property. By Lemma 3.1, there exists an ultrafilter p containing every $\mathbb{N} \setminus A_i$. Then no A_i belongs to p, so $p \notin \overline{A_i}$ for every $i \in I$, contradicting that the sets $\overline{A_i}$ cover $\beta\mathbb{N}$. Thus every basic clopen cover has a finite subcover, and $\beta\mathbb{N}$ is compact. □

For $A \subseteq \mathbb{N}$ and $n \in \mathbb{N}$, put

$$A - n := \{m \in \mathbb{N} : m + n \in A\}.$$

Addition on $\mathbb{N}$ extends to $\beta\mathbb{N}$ as follows:

$$A \in p + q \quad \Longleftrightarrow \quad \{n \in \mathbb{N} : A - n \in q\} \in p.$$

Proposition 3.3. *With the operation just defined, $\beta\mathbb{N}$ is a compact Hausdorff right-topological semigroup extending ordinary addition on principal ultrafilters.*

Proof. Compactness and Hausdorffness are Proposition 3.2. We first check that $p + q$ is an ultrafilter whenever $p, q \in \beta\mathbb{N}$.

Fix $q \in \beta\mathbb{N}$ and define

$$T_q(A) := \{n \in \mathbb{N} : A - n \in q\}.$$

The map T_q respects the empty set, the whole space, complements, finite intersections, and supersets:

$$T_q(\varnothing) = \varnothing, \qquad T_q(\mathbb{N}) = \mathbb{N}, \qquad T_q(\mathbb{N} \setminus A) = \mathbb{N} \setminus T_q(A), \qquad T_q(A \cap B) = T_q(A) \cap T_q(B),$$

and $A \subseteq B$ implies $T_q(A) \subseteq T_q(B)$. Therefore

$$p + q := \{A \subseteq \mathbb{N} : T_q(A) \in p\}$$

is an ultrafilter.

If p_m, p_n are principal ultrafilters at $m, n \in \mathbb{N}$, then

$$A \in p_m + p_n \Longleftrightarrow A - m \in p_n \Longleftrightarrow m + n \in A.$$

Thus $p_m + p_n = p_{m+n}$, so the operation extends ordinary addition.

Now we prove associativity. Let $p, q, r \in \beta\mathbb{N}$ and $A \subseteq \mathbb{N}$. Then

$$\begin{aligned} A \in (p + q) + r &\Longleftrightarrow \{n : A - n \in r\} \in p + q \\ &\Longleftrightarrow \{m : (\{n : A - n \in r\}) - m \in q\} \in p. \end{aligned}$$

But

$$(\{n : A - n \in r\}) - m = \{n : A - (m + n) \in r\} = \{n : (A - m) - n \in r\}.$$

Therefore

$$(\{n : A - n \in r\}) - m \in q \quad \Longleftrightarrow \quad A - m \in q + r.$$

Hence

$$\begin{aligned} A \in (p + q) + r &\Longleftrightarrow \{m : A - m \in q + r\} \in p \\ &\Longleftrightarrow A \in p + (q + r). \end{aligned}$$

Thus addition is associative.

Finally, fix $q \in \beta\mathbb{N}$. For every $A \subseteq \mathbb{N}$,

$$\{p \in \beta\mathbb{N} : A \in p + q\} = \overline{\{n \in \mathbb{N} : A - n \in q\}}.$$

This is clopen. Therefore the map $p \mapsto p + q$ is continuous. Thus $\beta\mathbb{N}$ is right-topological. □

Lemma 3.4 (Ellis–Numakura idempotent lemma)**.** *Every non-empty compact Hausdorff right-topological semigroup contains an idempotent element.*

Proof. Let S be a non-empty compact Hausdorff right-topological semigroup. We first choose a minimal non-empty compact subsemigroup. Partially order the non-empty compact subsemigroups of S by inclusion. If $\{K_i : i \in I\}$ is a descending chain of such subsemigroups, then every finite subfamily has non-empty intersection. Since compact subsets of a Hausdorff space are closed, compactness of S gives

$$\bigcap_{i \in I} K_i \neq \varnothing.$$

This intersection is again a compact subsemigroup. Hence Zorn's lemma gives a minimal non-empty compact subsemigroup; call it K.

Fix $a \in K$. Since right multiplication by a is continuous, the set

$$Ka := \{xa : x \in K\}$$

is compact and non-empty. It is also a subsemigroup: if $xa, ya \in Ka$, then by associativity

$$(xa)(ya) = x(ay)a,$$

and $x(ay) \in K$ because K is a subsemigroup. Thus $(xa)(ya) \in Ka$. By minimality of K, we have $Ka = K$.

Since $a \in K = Ka$, there exists $e \in K$ such that $ea = a$. Now define

$$E := \{x \in K : xa = a\}.$$

This set is non-empty because $e \in E$. It is compact because it is the inverse image of the closed set $\{a\}$ under the continuous right-translation map $x \mapsto xa$. It is also a subsemigroup: if $x, y \in E$, then

$$(xy)a = x(ya) = xa = a.$$

Therefore E is a non-empty compact subsemigroup of K. By minimality of K, $E = K$. In particular, $a \in E$, so

$$a^2 = a.$$

Thus a is an idempotent. □

Corollary 3.5. *There exists a non-principal idempotent ultrafilter $p \in \beta\mathbb{N}$; that is,*

$$p + p = p.$$

Proof. By Proposition 3.3, $\beta\mathbb{N}$ is a non-empty compact Hausdorff right-topological semigroup. Lemma 3.4 gives an idempotent ultrafilter p.

It cannot be principal. If p were the principal ultrafilter at $n \in \mathbb{N}$, then $p + p$ would be the principal ultrafilter at $2n$. Since $n \geq 1$, $2n \neq n$, so $p + p \neq p$. Hence p is non-principal. □

We shall use one elementary consequence of non-principality.

Lemma 3.6. *If p is a non-principal ultrafilter on $\mathbb{N}$ and $A \in p$, then A is infinite and therefore contains arbitrarily large integers.*

Proof. If a finite set E belonged to p, then, because

$$E = \bigcup_{n \in E} \{n\},$$

the ultrafilter property would force $\{n\} \in p$ for some $n \in E$. Indeed, if no singleton $\{n\}$ belonged to p, then each complement $\mathbb{N} \setminus \{n\}$ would belong to p, and intersecting these complements over $n \in E$ would show $\mathbb{N} \setminus E \in p$, contradicting $E \in p$.

But if $\{n\} \in p$, then for every $B \subseteq \mathbb{N}$, the ultrafilter property gives

$$B \in p \iff n \in B,$$

so p is principal at n. This contradicts the assumption. Therefore no finite set lies in p. Hence every $A \in p$ is infinite, and every infinite subset of $\mathbb{N}$ is unbounded. $\square$

3.2 Hindman's theorem

Theorem 3.7 (Hindman's finite-sums theorem)**.** *For every finite colouring*

$$\chi : \mathbb{N} \to \{1, \dots, r\},$$

there exists a strictly increasing sequence

$$x_1 < x_2 < x_3 < \cdots$$

of positive integers and a colour c such that every non-empty finite sum

$$x_{i_1} + \cdots + x_{i_m}, \qquad i_1 < \cdots < i_m,$$

has colour c.

Proof. Let $p \in \beta\mathbb{N}$ be a non-principal idempotent ultrafilter, whose existence is guaranteed by Corollary 3.5. Let the colour classes be

$$C_i := \chi^{-1}(i), \qquad i = 1, \dots, r.$$

Because p is an ultrafilter and

$$\mathbb{N} = C_1 \cup \cdots \cup C_r,$$

exactly one of the sets C_i belongs to p. Denote this set by C.

Define

$$C^* := \{n \in C : C - n \in p\}.$$

Since $p + p = p$ and $C \in p$, the definition of addition gives

$$D := \{n \in \mathbb{N} : C - n \in p\} \in p.$$

Thus

$$C^* = C \cap D \in p.$$

We now prove the fundamental shift property:

$$n \in C^* \implies C^* - n \in p. \tag{1}$$

Indeed, if $n \in C^*$, then $C - n \in p$. Also, since $C - n \in p = p + p$,

$$\{m : (C - n) - m \in p\} \in p.$$

But

$$(C - n) - m = C - (n + m).$$

Therefore

$$\{m : C - (n + m) \in p\} \in p.$$

This set is exactly $D - n$. Hence

$$C^* - n = (C - n) \cap (D - n) \in p.$$

This proves (1).

We construct $x_1, x_2, \dots$ recursively. Choose any $x_1 \in C^*$; this is possible because $C^* \in p$ and therefore C^* is non-empty.

Suppose $x_1 < \cdots < x_t$ have already been chosen so that every non-empty finite sum of these t integers lies in C^*. Let

$$\mathrm{FS}_t := \left\{ \sum_{i \in I} x_i : \varnothing \neq I \subseteq \{1, \dots, t\} \right\}.$$

For each $s \in \mathrm{FS}_t$, we have $s \in C^*$, so by (1),

$$C^* - s \in p.$$

Consequently

$$E_t := C^* \cap \bigcap_{s \in \mathrm{FS}_t} (C^* - s) \in p.$$

By Lemma 3.6, E_t contains arbitrarily large integers. Choose

$$x_{t+1} > x_t$$

with $x_{t+1} \in E_t$.

Then $x_{t+1} \in C^*$, and for every $s \in \mathrm{FS}_t$,

$$s + x_{t+1} \in C^*,$$

because $x_{t+1} \in C^* - s$. Therefore every non-empty finite sum from $x_1, \dots, x_{t+1}$ lies in C^*. By induction, this holds for all t.

Since $C^* \subseteq C$, every non-empty finite sum from the infinite sequence has the single colour corresponding to C. □

4 From Hindman's theorem to the finite number $F(k)$

We now prove the qualitative upper bound $F(k) < \infty$.

Lemma 4.1. *For every $k \geq 1$, there exists an integer N such that every two-colouring of $\{1, \dots, N\}$ contains a k-element set A with $\mathrm{FS}(A) \subseteq \{1, \dots, N\}$ and with $\mathrm{FS}(A)$ monochromatic.*

Proof. Suppose, for contradiction, that no such N exists for some fixed k. Then for every $N \geq 1$ there is a bad colouring

$$\chi_N : \{1, \dots, N\} \to \{0, 1\}$$

with no valid monochromatic k-term finite-sums set.

We first construct an infinite colouring

$$\chi : \mathbb{N} \to \{0, 1\}$$

such that every finite initial segment of χ agrees with some bad finite colouring of sufficiently large length.

Start with the infinite set $I_0 := \mathbb{N}$. At stage $m \geq 1$, assume I_{m-1} is an infinite set of indices N such that $N \geq m - 1$ and such that the colourings χ_N agree with the already chosen values $\chi(1), \dots, \chi(m-1)$ on $\{1, \dots, m-1\}$. Infinitely many elements of I_{m-1} are at least m. Among those N, the value $\chi_N(m)$ is either 0 or 1; one of these two values occurs for infinitely many such N. Define $\chi(m)$ to be that value, and let I_m be the infinite set of corresponding indices.

This recursive construction has the following consequence: for every $M \geq 1$, there exists some $N \geq M$ such that

$$\chi_N(n) = \chi(n) \qquad (1 \leq n \leq M). \tag{2}$$

Indeed, any $N \in I_M$ works.

Apply Hindman's theorem, Theorem 3.7, to the infinite colouring χ. There exist distinct positive integers

$$x_1 < x_2 < x_3 < \cdots$$

such that all non-empty finite sums from this sequence have the same colour. Put

$$A := \{x_1, \ldots, x_k\}$$

and

$$M := x_1 + \cdots + x_k.$$

Then $\mathrm{FS}(A) \subseteq \{1, \ldots, M\}$ and $\mathrm{FS}(A)$ is monochromatic under χ.

Choose $N \geq M$ such that χ_N agrees with χ on $\{1, \ldots, M\}$, as guaranteed by (2). Then $\mathrm{FS}(A)$ is also monochromatic under χ_N, and it is contained in $\{1, \ldots, N\}$. This contradicts the fact that χ_N was chosen to be bad.

Therefore such an N must exist. □

Corollary 4.2. *For every $k \geq 1$, $F(k) < \infty$.*

Proof. Lemma 4.1 shows that the set of integers N satisfying the defining property of $F(k)$ is non-empty. It is also upward closed: if N has the property and $M \geq N$, then any colouring of $\{1, \ldots, M\}$ restricts to a colouring of $\{1, \ldots, N\}$, where the desired set A already exists. Thus the least such N exists. □

5 Conclusion

Combining Corollary 2.2 and Corollary 4.2, we obtain Theorem 1.2:

$$4^{k-1} \leq F(k) < \infty.$$

The estimate has two different strengths. The lower bound is explicit and shows that $F(k)$ grows at least exponentially in k. The upper bound is qualitative: it proves that $F(k)$ is always finite, but it does not provide a manageable numerical upper bound. Quantitative estimates for finite-sums Ramsey numbers are substantially more delicate than ordinary Schur or Ramsey bounds.